

NMAP CHEAT SHEET

Tips for conducting a Nmap scan.

Basic Scanning Techniques

Scan a single target	nmap [target]
Scan multiple targets	nmap [target1,target2,etc]
Scan a list of targets	nmap -iL [list.txt]
Scan a range of hosts	nmap [range of IP addresses]
Scan an entire subnet	nmap [IP address/cdir]
Scan random hosts	nmap -iR [number]
Excluding targets from a scan	nmap [targets] –exclude [targets]
Excluding targets using a list	nmap [targets] –excludefile [list.txt]
Perform an aggressive scan	nmap -A [target]
Scan an IPv6 target	nmap -6 [target]

Discovery Options

Perform a ping scan only	nmap -sP [target]
Don't ping	nmap -PN [target]
TCP SYN Ping	nmap -PS [target]
TCP ACK ping	nmap -PA [target]
UDP ping	nmap -PU [target]
SCTP Init Ping	nmap -PY [target]
ICMP echo ping	nmap -PE [target]
ICMP Timestamp ping	nmap -PP [target]
ICMP address mask ping	nmap -PM [target]
IP protocol ping	nmap -PO [target]
ARP ping	nmap -PR [target]
Traceroute	nmap –traceroute [target]
Force reverse DNS resolution	nmap -R [target]
Disable reverse DNS resolution	nmap -n [target]

Advanced Scanning Options

TCP SYN Scan	nmap -sS [target]
TCP connect scan	nmap -sT [target]
UDP scan	nmap -sU [target]
TCP Null scan	nmap -sN [target]
TCP Fin scan	nmap -sF [target]
Xmas scan	nmap -sX [target]
TCP ACK scan	nmap -sA [target]
Custom TCP scan	nmap –scanflags [flags] [target]
IP protocol scan	nmap -sO [target]
Send Raw Ethernet packets	nmap –send-eth [target]
Send IP packets	nmap –send-ip [target]

Port Scanning Options

Perform a fast scan	nmap -F [target]
Scan specific ports	nmap -p [ports] [target]
Scan ports by name	nmap -p [port name] [target]
Scan ports by protocol	nmap -sU -sT -p U:[ports],T:[ports] [target]
Scan all ports	nmap -p “*” [target]
Scan top ports	nmap –top-ports [number] [target]
Perform a sequential port scan	nmap -r [target]

Version Detection

Operating system detection	nmap -O [target]
Attempt to guess an unknown	nmap -O –osscan-guess [target]
Service version detection	nmap -sV [target]
Troubleshooting version scans	nmap -sV –version-trace [target]
Perform a RPC scan	nmap -sR [target]

Timing Options

Timing Templates	nmap -T [0-5] [target]
Set the packet TTL	nmap –ttl [time] [target]
Minimum of parallel connections	nmap –min-parallelism [number] [target]
Maximum of parallel connection	nmap –max-parallelism [number] [target]
Minimum host group size	nmap –min-hostgroup [number] [targets]
Maximum host group size	nmap –max-hostgroup [number] [targets]
Maximum RTT timeout	nmap –initial-rtt-timeout [time] [target]
Initial RTT timeout	nmap –max-rtt-timeout [TTL] [target]
Maximum retries	nmap –max-retries [number] [target]
Host timeout	nmap –host-timeout [time] [target]
Minimum Scan delay	nmap –scan-delay [time] [target]
Maximum scan delay	nmap –max-scan-delay [time] [target]
Minimum packet rate	nmap –min-rate [number] [target]
Maximum packet rate	nmap –max-rate [number] [target]
Defeat reset rate limits	nmap –defeat-rst-ratelimit [target]

